



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Main)

SCORE-3.0

20-06-2021

JEE(Main + Advance) : Leader & Enthusiast Course (PHASE : SCORE-3.0)

Test Type : SCORE-3.0

Test Pattern : JEE (Main)

TEST DATE : 20-06-2021

ANSWER KEY

PAPER-2

PHYSICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	B	B	A	A	D	C	B	A
SEC-B	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	A	D	C	A	A	C	B	C	A

CHEMISTRY

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	B	C	D	A	B	C	B	D
SEC-B	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	A	A	B	C	C	A	C	B	A

MATHEMATICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	C	B	A	B	A	C	C	B
SEC-B	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	A	C	A	A	A	C	D	C

HINT – SHEET

PHYSICS

SEC-A

1. Ans (A)

Primary rainbow is between 40° & 42° above horizontal. So, its angular spread is 2° and for secondary rainbow angles are between 50° & 53° above horizontal. So, its angular spread is 3° .

Difference between both these angular spread is 1°

2. Ans (A)

For relaxed eye adjustment,

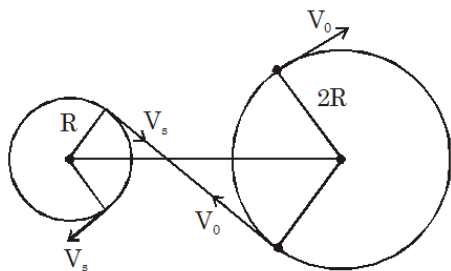
$$\text{mag.} = \frac{f_0}{f_e}$$

$$\text{So, } f_e = 10 \text{ cm}$$

and separation between lenses

$$= f_0 + f_e = 110 \text{ cm}$$

3. **Ans (B)**



$$f_{\max} = \left(\frac{C + V_0}{C - V_s} \right) f$$

$$f_{\min} = \left(\frac{C - V_0}{C + V_s} \right) f$$

$$\Delta f = f_{\max} - f_{\min} = \frac{2C(V_s + V_0)}{(C^2 - V_s^2)} f$$

$$\text{and } V_s = 2R \text{ \& } V_0 = \omega(2R)$$

$$\Delta f = \frac{6C(\omega R)f}{C^2 - (\omega R)^2}$$

4. **Ans (B)**

Intensity from a linear source

$$I = \frac{P}{2\pi r \ell} \propto \frac{1}{r}$$

Change in loudness level

$$\Delta L = 10 \log_{10} \left(\frac{I_1}{I_2} \right) \text{ dB}$$

$$= 10 \log_{10} \left(\frac{r_2}{r_1} \right) \text{ dB}$$

5. **Ans (A)**

Resolving power is given by the reciprocal of the minimum separation between two points seen as distinct

$$= \frac{D}{1.22\lambda L}$$

6. **Ans (A)**

Focal length of lens

$$\frac{1}{f} - (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$(\mu - 1) = \frac{1}{f \left(\frac{1}{R_1} - \frac{1}{R_2} \right)} = \frac{k}{f}$$

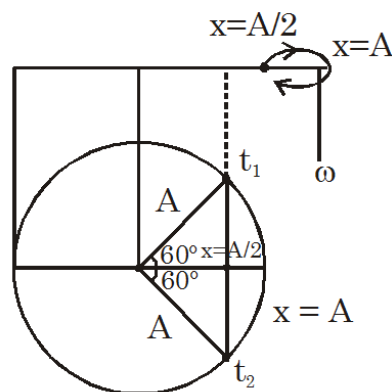
dispersive power

$$\omega = \frac{\mu_v - \mu_R}{\mu_y - 1}$$

$$= \frac{\left(\frac{k}{f_v} + 1 \right) - \left(\frac{k}{f_R} - 1 \right)}{\left(\frac{k}{f_y} \right)} = \frac{\left(\frac{1}{f_v} + \frac{1}{f_R} \right)}{\left(\frac{1}{f_y} \right)} \approx 0.041$$

7. **Ans (D)**

Phasor diagram



$$\text{Phase angle rotated} = 120^\circ = \frac{2\pi}{3}$$

$$\text{time taken} = \frac{2\pi}{3\omega} = \frac{T}{3}$$

8. **Ans (C)**

$$x = 3 \sin 100t + 8 \cos^2 50t$$

$$= 3 \sin 100t + 8 \left(\frac{1 + \cos 2\theta}{2} \right)$$

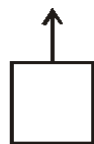
$$= 3 \sin 100t + 4 \cos 100t + 4$$

$$x = 5 \sin (100t) + 4$$

$$\Rightarrow A = 5, \text{ motion is } 5.17 \text{ M}$$

9. **Ans (B)**

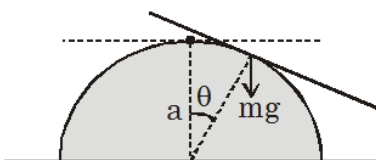
$$F_{\text{net}} = \rho A x g$$



$$F_{\text{net}} = \rho A x g \Rightarrow \frac{\rho}{3} A l g \times a = \rho A x g \Rightarrow a = \frac{3x}{l}$$

$$\text{time period} = 2\pi \sqrt{\frac{l}{3g}}$$

10. **Ans (A)**



$$I\alpha = t = mg \sin \theta$$

$$\frac{m(2\ell)^2}{12} \alpha = mga \theta$$

$$\alpha = \left(\frac{3ga}{\ell^2} \right) \theta = \omega^2 \theta$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi \ell}{\sqrt{3ga}}$$

12. **Ans (A)**

$$n = \frac{u}{4\ell_1} = \frac{pv}{2\ell_2}$$

$$p = \frac{2\ell_2}{4\ell_1} = \frac{2 \times 80}{4 \times 20}$$

13. **Ans (D)**

$$2A \sin kx = 3\sqrt{2}$$

$$2 \times 3 \sin kx = 3\sqrt{2}$$

$$\sin kx = \frac{1}{\sqrt{2}}$$

$$\frac{2\pi}{\lambda}x = \frac{\pi}{4}; \frac{3\pi}{4}$$

$$x = \frac{\lambda}{8}; \frac{3\lambda}{8} \dots\dots\dots$$

Distance between consecutive points

$$= \frac{3\lambda}{8} - \frac{\lambda}{8} = \frac{\lambda}{4}$$

$$\frac{\lambda}{4} = 20 \text{ cm}$$

$$\Rightarrow \lambda = 80 \text{ cm}$$

$$\text{So, } (n+1) \frac{\lambda}{2} = 240$$

$$\Rightarrow (n+1) \frac{80}{2} = 240$$

$$\text{or } n+1 = 6$$

$$n = 5$$

So string vibrating in fifth overtone.

$\therefore$ (D)

14. **Ans (C)**

$$f\lambda = \sqrt{\frac{\gamma RT}{M}}$$

$$1100 \times 0.4 = \sqrt{\frac{\gamma \times \frac{25}{3} \times 300}{16 \times 10^{-3}}}$$

$$\frac{4 \times 440 \times 440 \times 16 \times 10^{-3}}{4 \times 25} = \gamma$$

$$= 121 \times 1024 \times 10^{-5} = \gamma$$

$$1.24 = \gamma$$

15. **Ans (A)**

$$V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{10}{0.1}} = 10 \text{ m/s}$$

$$y = 2 \sin (20t - 2x)$$

$$\frac{\partial y}{\partial t} = v_p = 0.02 \cos(t - 2x)$$

Energy density

$$= \frac{I}{V} \times \text{Area} =$$

$$\frac{\partial \times A^2 \omega^2 \cos^2(\omega t - kx) \times \text{Area} \times v}{v}$$

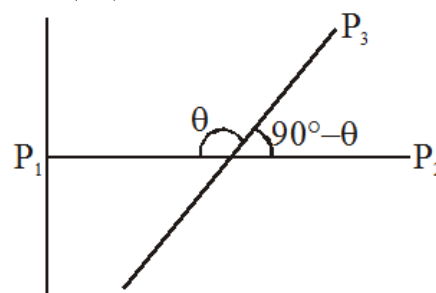
At near position,

$$\text{Energy density} = \mu \times A^2 \omega^2$$

$$= 0.1 \times 4 \times 10^{-4} \times 400$$

$$= 1.6 \times 10^{-2} \text{ J/m}$$

16. **Ans (A)**



Let initial intensity of light is I_0 . So intensity of light after transmission from first polaroid = $\frac{I_0}{2}$.

Intensity of light emitted from P_3

$$I_1 = \frac{I_0}{2} \cos^2 \theta$$

Intensity of light transmitted from last polaroid

$$P_2 = I_1 \cos^2 (90^\circ - \theta)$$

$$P_2 = \frac{I_0}{2} \cos^2 \theta \sin^2 \theta$$

$$P_2 = \frac{I_0}{8} (2 \sin \theta \cos \theta)^2$$

$$P_2 = \frac{I_0}{8} \sin^2 2\theta$$

17. **Ans (C)**

For small angles $\sin \theta \approx \theta$.

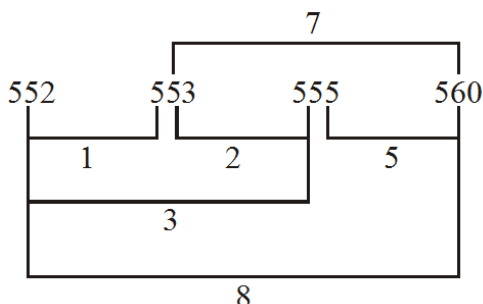
$$d \sin \theta = d\theta = \lambda$$

$$\Rightarrow \theta = \frac{\lambda}{d} \Rightarrow d = \frac{\lambda}{\theta}$$

$$\text{Since } d \text{ is constant } \frac{\lambda_1}{\theta_1} = \frac{\lambda_2}{\theta_2}$$

$$\therefore \lambda_2 = 720 \text{ nm}$$

18. Ans (B)



20. Ans (A)

Here, $E = 6\text{V/m}$, $c = 3 \times 10^8 \text{ ms}^{-1}$

$$B = \frac{E}{c} = \frac{6\text{V/m}}{3 \times 10^8 \text{ ms}^{-1}} = 2 \times 10^{-8} \text{ T}$$

E is along the y -direction and the plane e.m. wave propagate along x -direction. Therefore, B should be in a direction perpendicular to both x and y -axis. Using vector algebra $\vec{E} \times \vec{B}$ should be along x -direction. Since $(+\hat{j}) \times (+\hat{k}) = \hat{i}$, B is along the z -direction.

Thus, magnetic field component B would be $2 \times 10^{-8} \text{ T}$ along z -direction.

PHYSICS

SEC-B

1. Ans (20.00)

For lens,

$$u = -15 \text{ cm}$$

$$f = +20 \text{ cm}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$v = -60 \text{ cm}$$

This image position will act as object for mirror. To coincide final image with object, the light rays must retrace their path for which object for mirror must be at its center of curvature

hence,

$$|v| + d = R$$

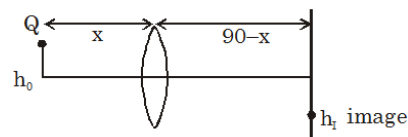
$$60 + d = 80$$

$$d = 20 \text{ cm}$$

2. Ans (21.60)

$$h_1 h_2 = h_0^2$$

$$\Rightarrow h_0 = 6 \text{ mm}$$



$$\frac{h_I}{h_0} = \frac{d_I}{d_0}$$

$$\frac{9\text{mm}}{6\text{mm}} = \frac{(90-x)\text{cm}}{(x)\text{cm}}$$

$$x = 36 \text{ cm}$$

$$90 - x = 54 \text{ cm}$$

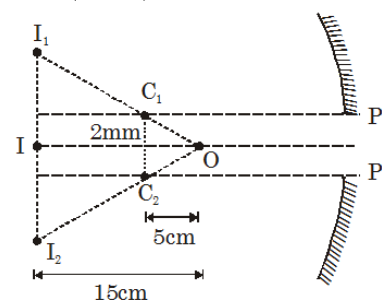
Lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{54} - \frac{1}{-36} = \frac{1}{f}$$

$$f = 21.6 \text{ cm}$$

3. Ans (6.00)



For mirror

$$u = -15 \text{ cm}, f = -10\text{cm}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow v = -30\text{cm}$$

C_1 & C_2 are positions of center of curvatures of the two halves, and line joining object and image passes through center of curvature using similar Δ s, $I_1 I_2$ separation = 6mm.

4. Ans (1.16 to 1.17)

$$T = 2\pi \sqrt{\left(\frac{M + \frac{m_s}{3}}{k}\right)}$$

$$3 = 2\pi \sqrt{\left(\frac{M + \frac{m_s}{3}}{k}\right)}$$

$$2 = 2\pi \sqrt{\left(\frac{M + \frac{m_s/2}{3}}{2k}\right)}$$

Divide and solve for $\frac{M}{m_s}$,

$$\text{We get } \frac{M}{m_s} = \frac{7}{6}$$

5. Ans (150.00)

Amplitude $A = 20 \text{ mm} = 2 \times 10^{-2} \text{ m}$

When, PE is maximum, KE = minimum

$= 0.04 \text{ J} = 0$

$TE = PE + KE$

$= 0.04 \text{ J} + 0 = 0.04 \text{ J}$

and when PE is minimum,

$= 0.01 \text{ J}$

KE = maximum

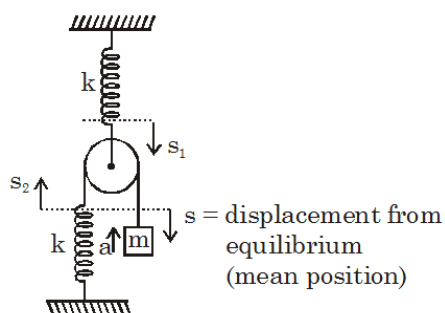
$= TE - PE_{\min}$

$= 0.03 \text{ J}$

and $KE_{\max} = \frac{1}{2}mv_{\max}^2 = \frac{1}{2}m\omega^2 A^2$

$\Rightarrow 0.03 \text{ J} = \frac{1}{2}kA^2 \Rightarrow k = 150 \text{ N/m}$

6. Ans (5.00)



$T = ks_2$

and $2T = ks_1$

$\Rightarrow 2s_2 = s_1$

also by constraint motion

$s - s_2 = 2s_1$

$s = 2s_1 + s_2$

$= 2(2s_2) + s_2$

$= 5s_2$

$s_2 = \frac{8}{5}$

$a = \frac{T}{m} = \frac{ks_2}{m} = \frac{k(s/5)}{m}$

$a = \left(\frac{k}{5m} \right) s$

for SHM

$a = (\omega^2)s$

$\Rightarrow \omega = \sqrt{\frac{k}{5m}}$

Time period $= \frac{2\pi}{\omega}$

$= 2\pi\sqrt{\frac{5m}{k}}$

7. Ans (1.33 to 1.34)

Time period for half part :

$T = 2\pi\sqrt{\frac{\ell}{g}} = 2\pi\sqrt{\frac{1}{g}} = \frac{2\pi}{\pi} = 2 \text{ sec.}$

So 2° part will be covered in a time

$t = \frac{T}{2} = 1 \text{ sec.}$

For the left 1° part : $\theta = \theta_0 \sin(\omega t)$

$\Rightarrow 1^\circ = 2^\circ \sin\left(\frac{2\pi}{T} \times t\right)$

$\Rightarrow \frac{1}{2} = \sin\left(\frac{2\pi}{2} \times t\right)$

$\Rightarrow \frac{\pi}{6} = \pi \times t \Rightarrow t = 1/6 \text{ sec.}$

Total time $= \frac{T}{2} + 2t$

$\Rightarrow 1 + 2 \times \frac{1}{6} = 1 + \frac{1}{3} = \frac{4}{3} \text{ sec.}$

8. Ans (3.00)

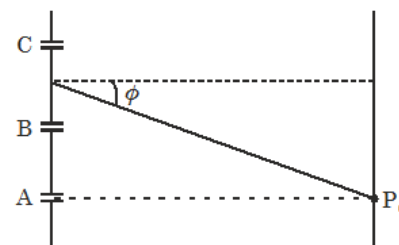
$\phi_{A/B}$ = Phase difference between A and B =

$\frac{2\pi}{\lambda} \times \frac{A}{3} = \frac{2\pi}{3}$

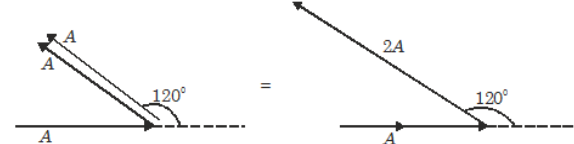
$\Delta x_{B/C}$ = path difference between B and C

$= d \sin \phi = d \tan \phi = \frac{3d^2}{D} = \lambda$

$\therefore \phi_{B/C}$ = phase difference between B and C $= 2\pi$



Phase diagram of waves arriving at P_0 in as shown



Resultant amplitude is given by

$A' = \sqrt{A^2 + (2A)^2 + 2(A)(2A)\cos 120^\circ} = \sqrt{3}A$

As $I \propto a^2 \therefore I' = 3I$

9. Ans (0.88 to 0.89)

$A_r = \left(\frac{v_1 - v_2}{v_1 + v_2} \right) A_i ; v_2 = \frac{v_1}{2}$

$\therefore \text{energy reflected} = \frac{1}{9} \times \text{energy incident}$

10. Ans (200.00)

$$\frac{1}{2L} \times v + 100 = \frac{3}{4L} v$$

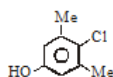
$$100 = \frac{3v}{4L} - \frac{2v}{4L} = \frac{v}{4L} \Rightarrow \frac{v}{2L} = 200 \text{ Hz}$$

CHEMISTRY

SEC-A

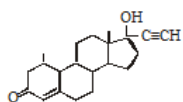
15. Ans (C)

(A) Chloroxylenol



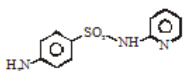
FeCl₃
test

(B) Norethindrone



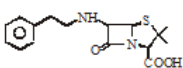
Bayer's
test

(C) Sulphapyridine



Carbylamine
test

(D) Penicillin



Sodium hydrogen
carbonate test

MATHEMATICS

SEC-A

1. Ans (C)

$$\lim_{n \rightarrow \infty} \left(\left(1 + \frac{1}{n}\right) \left(1 + \frac{1}{2n}\right) \left(1 + \frac{1}{2^2 n}\right) \dots \left(1 + \frac{1}{2^{n-1} n}\right) \right)^n$$

$$= e^{1 + \frac{1}{2} + \frac{1}{2^2} + \dots \infty} = e^2$$

2. Ans (C)

$$\lim_{x \rightarrow 0} x e^{-\left(\frac{1}{|x|} + \frac{1}{x}\right)} = 0$$

and $f(0) = 0$

$\Rightarrow y = f(x)$ is continuous at $x = 0$

now at $x = 0$

$$\text{L.H.D.} = \lim_{h \rightarrow 0} \frac{-h \cdot e^0 - 0}{-h} = 1$$

$$\text{R.H.D.} = \lim_{h \rightarrow 0} \frac{h \cdot e^{-2/h} - 0}{h} = 0$$

$\Rightarrow$ Not differentiable

3. Ans (C)

$$\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\left(\frac{\sin x}{x - \sin x} \right)}$$

$$= e^{\lim_{x \rightarrow 0} \left(\frac{\sin x - x}{x} \right) \cdot \left(\frac{\sin x}{x - \sin x} \right)} = e^{-1}$$

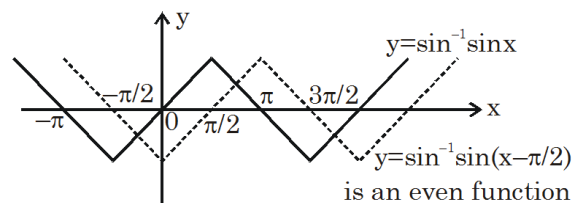
4. Ans (B)

Area of circles $= n^2 \cdot \pi r^2$

and area of square $= (2nr)^2$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{\pi n^2 r^2}{4n^2 r^2} = \frac{\pi}{4}$$

5. Ans (A)



$\Rightarrow y = \left[\sin^{-1} \sin \left(x - \frac{\pi}{2} \right) \right] + \frac{\pi}{2}$ will be an even function (symmetric in opposite quadrants)

6. Ans (B)

Let

$$g(x) = \{x\} + \left\{x + \frac{1}{n}\right\} + \left\{x + \frac{2}{n}\right\} + \dots + \left\{x + \frac{n-1}{n}\right\}$$

and

$$g\left(x + \frac{1}{n}\right) = \left\{x + \frac{1}{n}\right\} + \left\{x + \frac{2}{n}\right\} + \dots + \{x\}$$

$$\Rightarrow g(x) = g\left(x + \frac{1}{n}\right) \text{ period of } g(x) \text{ is } \frac{1}{n}$$

$$\Rightarrow g\left(\frac{x}{n}\right) = f(x) \Rightarrow \text{period of } f(x) = 1$$

7. Ans (A)

$$x^2 - 2x + 5 = (x-1)^2 + 4 \in [4, 13]$$

for $x \in [-2, 4]$

$y = [\log_2(x^2 - 2x + 5)] \in \{2, 3\}$ is continuous

for $\log_2(x^2 - 2x + 5) = 2$

but discontinuous for $\log_2(x^2 - 2x + 5) = 3$

$\Rightarrow$ 2 values of x

8. Ans (C)

$f(x) =$

$$\sec^{-1} \left(\frac{2x}{1+x^2} \right) + 2 \tan^{-1} x + \sin^{-1} (x^2 - 1)$$

$$\text{here } -1 \leq \frac{2x}{1+x^2} \leq 1$$

$$\Rightarrow \text{so } \sec^{-1} \text{ is defined only for } \frac{2x}{1+x^2} = \pm 1$$

$$\Rightarrow x = \pm 1$$

9. **Ans (C)**

$$f(x) = \cos^{-1} \left(\frac{1 - (\log_e x)^2}{1 + (\log_e x)^2} \right)$$

$$= 2\tan^{-1}(\log_e x) \quad \text{for } x \geq 1$$

$$\frac{dy}{dx} = \frac{2 \times \frac{1}{x}}{1 + (\log_e x)^2}$$

$$\Rightarrow f'(e) = \frac{1}{e}$$

10. **Ans (B)**

$$-2 \leq \alpha, \beta \leq 2$$

$$\Rightarrow f(-2) \geq 0, f(2) \geq 0, D \geq 0, \frac{-b}{2a} \in [-2, 2]$$

11. **Ans (A)**

$$\underbrace{(\cos^{-1} x - 2\pi)}_{\text{always } (-ve)} (x^2 - 1) \leq 0$$

$$\Rightarrow x^2 - 1 \geq 0 \text{ but according to } \cos^{-1} x, x \in [-1, 1]$$

$$x \in (-\infty, -1] \cup [1, \infty)$$

12. **Ans (B)**

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \tan^{-1} \frac{3}{9r^2 - 3r - 1}$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \tan^{-1} \frac{(3r+1) - (3r-2)}{1 + (3r+1)(3r-2)}$$

13. **Ans (A)**

$$\sin 2x(4(1 - \sin^2 x) - 1)(1 - 4(1 - \cos^2 x)) \geq 0$$

$$2\sin x \cos x(3 - 4\sin^2 x)(4\cos^2 x - 3) \geq 0$$

$$2(\sin 3x)(\cos 3x) \geq 0$$

$$\sin 6x \geq 0$$

14. **Ans (C)**

$$2x + y < 50$$

$$\text{if } y = 1 \Rightarrow x \in \{1, 2, \dots, 24\}$$

$$\text{if } y = 2 \Rightarrow x \in \{1, 2, \dots, 23\}$$

$$\text{if } y = 3 \Rightarrow x \in \{1, 2, \dots, 23\}$$

$$\text{if } y = 4 \Rightarrow x \in \{1, 2, \dots, 22\}$$

$$\text{if } y = 5 \Rightarrow x \in \{1, 2, \dots, 22\}$$

$$\text{if } y = 6 \Rightarrow x \in \{1, 2, \dots, 21\}$$

$$\vdots$$

$$\vdots$$

$$\text{if } y = 46 \Rightarrow x \in \{1\}$$

$$\text{if } y = 47 \Rightarrow x \in \{1\}$$

$$\text{total integral points} = 2[1 + 2 + 3 + \dots + 23] + 24$$

$$= 2 \times \frac{23 \times 24}{2} + 24 = 576$$

15. **Ans (A)**

$$\text{Use } \frac{3 \sin \theta - \sin 3\theta}{4} = \sin^3 \theta$$

$$\text{then } \sin \alpha + \sin(\alpha + \beta) + \dots + \sin(\alpha + (n-1)\beta)$$

$$= \frac{\sin \frac{n\beta}{2} \sin \left(\alpha + \frac{(n-1)\beta}{2} \right)}{\sin \frac{\beta}{2}}$$

16. **Ans (A)**

$$32\cot\theta - [\cot\theta - \tan\theta] + 2\tan 2\theta + 4\tan 4\theta +$$

$$8\tan 8\theta + 16\cot 16\theta$$

$$= 32\cot\theta - 2[\cot 2\theta - \tan 2\theta] + 4\tan 4\theta + 8\tan 8\theta$$

$$+ 16\cot 16\theta$$

$$= 32\cot\theta - 4[\cot 4\theta - \tan 4\theta] + 8\tan 8\theta +$$

$$16\cot 16\theta$$

$$(\text{continue by using } \cot\theta - \tan\theta = 2\cot 2\theta)$$

17. **Ans (A)**

$$y = \log_5[(18\sin x - 24\sin^3 x + 24\cos x -$$

$$32\cos^3 x)^2 + 25]$$

$$y = \log_5[(6\sin 3x - 8\cos 3x)^2 + 25]$$

$$-10 \leq 6\sin x - 8\cos 3x \leq 10$$

$$\Rightarrow y \in [2, 3]$$

18. **Ans (C)**

$$\sec^2 x \in [1, \infty) - \{2\}$$

$$\sec^2 x - 2 \in [-1, \infty) - \{0\}$$

$$(\sec^2 x - 2)^2 \in (0, \infty)$$

$$\frac{9}{(\sec^2 x - 2)^2} \in (0, \infty) \text{ onto}$$

19. **Ans (D)**

$$(1) \text{ Let } y = \tan x \Rightarrow \frac{dy}{dx} = \sec^2 x > 0$$

$$\Rightarrow \text{but } \tan x \text{ is not one-one}$$

$$(3) \text{ It is true if the function is not defined piecewise}$$

$$(4) \text{ Always true}$$

20. **Ans (C)**

$$\text{Use graph}$$

$$\text{OR}$$

$$\text{Let } 0 \leq \alpha, \beta, \gamma, \delta \leq \pi$$

$$\text{such that } \sec \alpha = 3, \sec \beta = 4, \sec \gamma = -3, \sec \delta = -4$$

$$\text{then we can say } \alpha < \beta < \delta < \gamma$$

MATHEMATICS

SEC-B

1. **Ans (3.00)**

Using L'hopital

$$\lim_{h \rightarrow 0} \frac{[f'(2h + 2 + h^2)] \cdot (2 + 2h)}{[f'(h - h^2 + 1)] \cdot (1 - 2h)}$$

2. **Ans (0.50)**

$$\lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n r \cdot (n - r + 1)}{\sum_{r=1}^n r^2} = \lim_{n \rightarrow \infty} \frac{-\sum_{r=1}^n r^2 + (n+1) \sum_{r=1}^n r}{\sum_{r=1}^n r^2}$$

$$= \lim_{n \rightarrow \infty} -1 + \frac{(n+1) \cdot \frac{n(n+1)}{2}}{\frac{n(n+1)(2n+1)}{6}}$$

divided numerator & denominator by n^3
 $= \frac{1}{2}$

3. **Ans (3.00)**

$$y = [\cos^{-1} \cos x] \in \{0, 1, 2, 3\}$$

at $x = 0$ it is continuous but $x \in \{1, 2, 3\}$ it is discontinuous.

4. **Ans (3.46)**

$$y = \sin^{-1} (2x\sqrt{1-x^2}) + \sin^{-1} x$$

$$\frac{dy}{dx} = \frac{3}{\sqrt{1-x^2}}$$

5. **Ans (0.00)**

$$y = \cos^{-1} \left(\frac{x-1}{x+1} \right) + \sin^{-1} \left(\frac{x-1}{x+1} \right) = \frac{\pi}{2}$$

$$\Rightarrow \frac{dy}{dx} = 0$$

6. **Ans (0.03)**

$$\lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n r^2 \cdot (2r+1)^3}{\sum_{r=1}^n r \cdot (4r-1)^4} = \lim_{n \rightarrow \infty} \frac{2^3 \sum_{r=1}^n r^5}{4^4 \sum_{r=1}^n r^5}$$

$$= \frac{2^3}{4^4} = \frac{1}{32}$$

7. **Ans (2.00)**

$$x^2 - 2 = -\sqrt{x+2}$$

use graph.

8. **Ans (2.33, 2.34)**

$$|x^2 - 3x + 2| + |x^2 - 7x + 12| = |4x - 10|$$

$$\Rightarrow (x^2 - 3x + 2)(x^2 - 7x + 12) \leq 0$$

$$\Rightarrow x \in [1, 2] \cup [3, 4]$$

$$\Rightarrow a = 1, b = 2, c = 3, d = 4$$

$$\Rightarrow \frac{c+d}{a+b} = \frac{7}{3} = 2.33 \text{ or } 2.34$$

9. **Ans (1.50)**

$$\text{Let } S_n = 1 + 4 + 10 + 20 + 35 + 56 + \dots + T_n$$

$$S_n = 1 + 4 + 10 + 20 + 35 + \dots + T_n$$

$$T_n = 1 + 3 + 6 + 10 + 15 + 21 + \dots + t_n$$

$$T_n = 1 + 3 + 6 + 10 + 15 + \dots + t_n$$

$$t_n = 1 + 2 + 3 + 4 + \dots n \text{ terms}$$

$$t_n = \frac{n(n+1)}{2}$$

$$\Rightarrow T_n = \sum_{r=1}^n t_r = \frac{1}{2} \sum_{r=1}^n r^2 + r$$

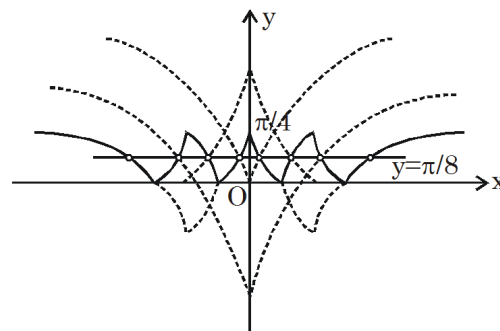
$$\Rightarrow T_n = \frac{n(n+1)(n+2)}{6}$$

Now

$$1 + \frac{1}{4} + \frac{1}{10} + \dots = \sum_{n=1}^{\infty} \frac{6}{n(n+1)(n+2)}$$

$$= 3 \sum_{n=1}^{\infty} \left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right) = \frac{3}{2}$$

10. **Ans (8.00)**



8 solutions